

BD Custom PC Builds — Research Task

Spec, source, and price 3 new desktop PCs to match our existing BD builds.

Assigned to: Jack **From:** Alex Black **Quantity:** 3 PCs

Reference build: BD Custom PC (built early 2025, ~\$1,685 ea) **Tooling:** claude.ai (research + cross-check)

1. The Job, In One Paragraph

We need 3 more desktop PCs at the same performance tier and aesthetic as the ones we built for the BD team. Your job is to research current 2026 parts, build a complete, cross-checked, compatibility-verified part list, and present a proposal with prices and links. You won't be assembling — just producing the research and proposal so we can place the order. Use **claude.ai** as a research partner the entire way; the back-and-forth is the point of the exercise.

2. Why These PCs Exist (Context You Need)

These desktops support the BD team's Power BI, Python data processing, and ML work. The original builds were spec'd around three priorities — keep these in mind because every part downstream serves them:

- **Memory-bound workloads.** Complex Power BI reports thrash memory and L3 cache. That's why we chose AMD's **X3D** chips — the extra 96 MB of 3D V-Cache is a massive win for PBI's repetitive data retrieval. There is no Intel equivalent for this specific workload pattern.
- **64 GB RAM floor.** Non-negotiable. Python dataframes and PBI models eat RAM.
- **Aesthetic consistency.** These sit on desks next to each other. The case must match the existing builds.

3. Hard Constraints (Don't Negotiate These)

Constraint	Detail
CASE — LOCKED	Same model as existing builds: Cooler Master ATX Mid Tower (the exact SKU we used — see Reference Build below). If unavailable new, the closest visual match from Cooler Master's current lineup, and flag this for approval before finalizing.
RAM — LOCKED	64 GB DDR5 minimum, per PC. Two-DIMM kit preferred (leaves slots for future upgrade).
CPU FAMILY — LOCKED	AMD X3D processor (Ryzen 7 7800X3D or its current-gen successor, e.g. 9800X3D). The 3D V-Cache is the whole reason. No Intel, no non-X3D AMD.
EVERYTHING ELSE	Negotiable on brand/model as long as performance tier and compatibility hold. See §5 for the reference parts to anchor against.

4. Reference Build (What We Used Before)

This is the existing BD PC. Treat it as the baseline to match or improve. Per-PC cost was ~**\$1,685** in early 2025.

Component	Part	Was	Status
Case	Cooler Master ATX Mid Tower	\$95	MATCH
CPU	AMD Ryzen 7 7800X3D (8C/16T)	\$497	X3D required
Motherboard	GIGABYTE B650 AORUS Elite AX V2 (AM5, DDR5, WiFi 6E)	\$210	tier match
CPU Cooler	Noctua NH-D15	\$120	tier match
RAM	Corsair Vengeance 64 GB DDR5-5200	\$140	64 GB floor
PSU	Corsair RM750x (ATX 3.1, 80+ Gold)	\$120	tier match
SSD (OS)	Samsung 1 TB NVMe	\$80	tier match
SSD (Data)	Samsung 2 TB NVMe	\$125	tier match
GPU	NVIDIA GeForce RTX 3060 12 GB	\$298	tier match (or current equiv)
Per-PC total		\$1,685	
× 3 PCs		\$5,055	

Full original list with links: [D:\PROJECTS\2024 Projects\PC Building\BD PC Part List Final.xlsx](#)

5. Step-by-Step Process

1 Inspect an existing BD PC in person.

- Open one of the existing units. Photograph the case (inside + outside, front + side), confirm the exact Cooler Master case model from the sticker or invoice.
- Confirm whether front I/O, fans, glass panel, and overall look matches what you see on Amazon/Newegg listings today.
- Note anything in the build you'd want to change (cable mess, fan noise, dust filter wear, etc.).

2 Start a claude.ai conversation. Prime it with context.

Paste in: the reference build table above, the three hard constraints, and the workload context (PBI / Python / ML). Tell Claude you're researching 3 new builds in **May 2026** at a similar tier. A good first message ends with: *"Before recommending parts, ask me any clarifying questions you need."*

Why this matters: Claude will give better answers if it knows you care about X3D L3 cache, 64 GB RAM, and case aesthetic consistency. Don't just ask "what PC should I build" — that gets you generic answers.

3 Research each component, one at a time.

For each row of the reference build, ask Claude:

- "Is this part still sold new in May 2026? If not, what's the closest current replacement?"
- "What's the current-generation equivalent at the same performance tier?"
- "What's the price range right now from Amazon, Newegg, Micro Center, B&H?"
- "Are there any known issues, recalls, or compatibility gotchas with this part?"

For the **CPU specifically**, ask Claude to compare the 7800X3D against the 9800X3D (and any newer X3D chip released by mid-2026): price, performance for Power BI / Python workloads, platform cost (motherboard + cooler compatibility). Recommend one and explain the tradeoff.

4 Verify every compatibility pair.

This is the part where mistakes get expensive. For your proposed list, confirm:

- **CPU ↔ Motherboard:** socket match (AM5), BIOS supports the CPU out of the box (or board has BIOS Flashback so we can update without a CPU).
- **RAM ↔ Motherboard:** on the QVL (qualified vendor list) ideally, or at least same DDR5 generation and speed the board officially supports.
- **CPU Cooler ↔ Case:** max CPU cooler height vs. case clearance. The Noctua NH-D15 is ~165 mm tall — borderline in many cases. Air vs. AIO is a real decision here.
- **GPU ↔ Case:** max GPU length vs. case clearance.
- **PSU ↔ everything:** wattage with ~30% headroom, correct connectors for the chosen GPU (12V-2x6 for newer NVIDIA).
- **SSDs ↔ Motherboard:** M.2 slot count and PCIe gen.

Tool: Run your full list through **PCPartPicker.com** — it auto-flags many compatibility issues. **Do not skip this.** It's the single best safety net.

5 Cross-check Claude with a second source.

Important. Claude is great for synthesis and tradeoff analysis but can be wrong about specific prices, current availability, and brand-new SKUs. **For every part you propose, open the product page and confirm:** (1) it's in stock, (2) the price Claude quoted is within ~10% of the live price, (3) the model number matches exactly. If Claude is off, update your list and tell Claude it was wrong — that helps the rest of the conversation stay accurate.

6 Build the proposal (the deliverable).

One spreadsheet, same shape as the original (BD PC Part List Final.xlsx) plus a short written summary. See §6 below for the exact deliverable.

6. Deliverable

Hand back two files:

A. Updated part list spreadsheet

Copy `BD PC Part List Final.xlsx`, save it as `BD PC Part List 2026 Proposal.xlsx`, and fill in the new columns. Each row should have:

- Component
- Proposed part (full model name + SKU)
- Purchase link (preferred vendor)
- Unit price (and date checked)
- Quantity (3)
- Line total
- "Same as 2025 build?" — Y / N / changed-because-X

B. One-page written summary

A short doc (PDF or Word, ≤ 1 page) covering:

- **Total cost** per PC and across 3 PCs, vs. the \$1,685 / \$5,055 baseline.
- **What changed and why.** For every component you swapped from the reference build, one sentence on why.
- **Case status.** Confirmed identical match — or, if not available, your recommended substitute with a side-by-side photo.
- **CPU recommendation.** 7800X3D vs. 9800X3D (or newest X3D) — which one and why, given price and workload.
- **Risks / open questions.** Anything you weren't sure about, anything I need to decide before we order.

7. Sourcing & Budget

Where to source

- Amazon (primary — we have business account)
- Newegg (often cheaper on GPUs/CPUs)
- Micro Center (in-person CPU + mobo bundles)
- B&H Photo (business-friendly)

Note vendor + price for each part. If a non-Amazon vendor saves $> \$30$ /part it's worth listing.

Budget

No hard cap. Anchor near \$1,685/PC at same performance tier. If 2026 prices push above that, explain the delta in your summary. If you can come in under at the same tier, even better.

8. How to Get the Most Out of claude.ai

- **One long conversation, not many short ones.** Keep all your research in a single Claude chat so context compounds.

- **Paste real data.** When you find a product page, paste the specs into Claude and ask "does this satisfy my constraints?" — that's more reliable than asking from memory.
- **Ask for tradeoffs, not just recommendations.** "Give me 3 options at different price points with the tradeoff between them" beats "what's the best motherboard."
- **Push back.** If Claude says X, ask "why not Y instead?" The reasoning it gives you is often the actual decision-making material.
- **Double-check pricing and availability against the actual product page.** Every time.

9. Timeline

Milestone	Target
Inspect existing PC, confirm case model	End of day 1
Draft part list + initial pricing	End of day 3
Compatibility check (PCPartPicker + manual)	End of day 4
Final proposal (spreadsheet + 1-pager) to Alex	End of week 1

10. Who to Ask

- **Spec questions / project context:** Alex Black
- **Anything about the existing builds:** Alex Black
- **IT / security / network integration:** not your concern for this task — that conversation is happening separately. Just produce the proposal.

Reference files: `D:\PROJECTS\2024 Projects\PC Building\BD PC Part List Final.xlsx` · invoices PC Invoice 2–9 · motherboard manual (B650 AORUS Elite AX V2). Skim the invoices to see exact SKUs we ordered.